Therefore, the probability for the occurrence of A given that B has occurred is

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{3}{9}}{\frac{4}{9}}$$

$$P(A/B) = \frac{3}{4}.$$

The probability for the occurrence of A given that C has occurred is

$$P(A/C) = \frac{P(A \cap C)}{P(C)} = \frac{\frac{2}{9}}{\frac{4}{9}}$$

$$P(A/C) = 1/2.$$

Similarly, the conditional probability of B given C is

$$P(B/C) = \frac{P(B \cap C)}{P(C)} = \frac{\frac{2}{9}}{\frac{4}{9}}$$

$$P(B/C) = \frac{1}{2}$$

and the conditional probability of B given A is

$$P(B/A) = \frac{P(B \cap A)}{P(A)} = \frac{\frac{3}{9}}{\frac{5}{9}}$$

$$P(B/A) = \frac{3}{5}.$$

Example 8.15

A pair of dice is rolled and the faces are noted. Let

A: sum of the faces is odd, B: sum of the faces exceeds 8, and

C: the faces are different then find (i) P (A/C) (ii) P (B/C)

Solution:

The outcomes favourable to the occurrence of these events are

A = { (1,2), (1,4), (1,6), (2,1), (2,3), (2,5), (3,2), (3,4), (3,6)
(4,1), (4,3), (4,5), (5,2), (5,4), (5,6), (6,1), (6,3), (6,5) }

B = { (3,6), (4,5), (4,6), (5,4), (5,5), (5,6), (6,3), (6,4), (6,5), (6,6) }

C = { (1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (2,3), (2,4), (2,5), (2,6),
(3,1), (3,2), (3,4), (3,5), (3,6), (4,1), (4,2), (4,3), (4,5), (4,6),
(5,1), (5,2), (5,3), (5,4), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5) }

Since A and B are proper subsets of C, $A \cap C = A$ and $B \cap C = B$.